Project Progress Report

Graded

1 Day, 19 Hours Late

Group

Venkata Satya Pradeep Srikakolapu
Sunil Mannuru
Sharath Chandra Kashetty
View or edit group

Total Points

190 / 200 pts

Question 1

Grader A (Select All)

94 / 100 pts

Proposed Method

- **– 0 pts** Satisfactory Description

List of Innovations

- **– 2 pts** 2 or more innovations mentioned, but the descriptions of each are vague.
- **– 0 pts** Unclear what is novel about listed innovation.
  See Submission Specific Adjustments [-2 per unclear innovation]

Experiments/Evaluation

- **– 0 pts** Satisfactory Description

Plan of Activities

- **– 0 pts** Satisfactory Plan of Activities

– 4 pts Good progress and interesting topic. See notes below:
- Significantly reduce the number of bullet point lists. Instead, make this read like a typical research paper. We will deduct points for this in final deliverables, so be sure to make these adjustments. It also makes it challenging to follow at times.
- Innovations section is vague. Please remove bullet points and write these in complete sentences. The novelty of using SHAP and clustering seem suspect.
- Your results for the LightGBM, XGBoost, etc models should be in the evaluation section, not in the methods. I suggest putting these in a data table and not in a bullet point list. Same with the SHAP results. You can introduce SHAP and the value it brings in methods, but save the results for the evaluation section.
- Why does logisitic regression not have precision and recall as well?
- Remove bullet points in evaluation section.
- You don't need to answer the HQ anymore. You can remove those in the final deliverables.
- I would adjust how you view the conclusion. Use this space to talk about how to interpret your results and what they mean to your application domain.

Question 2

Grader B (Select All)

96 / 100 pts

Proposed Method

- ✔ **– 0 pts** Satisfactory Description

List of Innovations

- ✔ **– 2 pts** 2 or more innovations mentioned, but the descriptions of each are vague.
- ✔ **– 0 pts** Unclear what is novel about listed innovation.
  See Submission Specific Adjustments [-2 per unclear innovation]

Experiments/Evaluation

- ✔ **– 0 pts** Satisfactory Description

Plan of Activities

- ✔ **– 0 pts** There are the old and revised plan of activities, and it mentions the time estimates and the responsibilities for each group member.

- **– 2 pts** Great project idea and good work on the project. I would strongly suggest changing your writing style from bullet points to a more research paper, full paragraph prose. Your methodology is clear, however I think some more depth could be provided regarding the novelty of your innovations. Excited to see where you take this project!

Questions assigned to the following page: 1 and 2

Interactive Drug Adverse Event Analysis and Prediction Platform

Progress Report — Team 217

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CSE 6242 — Data and Visual Analytics, Spring 2026, Georgia Institute of Technology

1. Introduction

Adverse drug events (ADEs) cause over 1.3 million deaths per year worldwide (WHO). The FDA Adverse Event Reporting System (FAERS) has 30M+ reports spanning 20+ years, but current tools—OpenVigil, VigiBase—are limited to basic statistics and static tables. We are building an interactive platform combining graph analytics, ML, and visualization for drug safety exploration. Our goals:

- Find hidden drug–side-effect associations using network analysis.
- Detect drug–drug interactions not yet widely recognized, via community detection.
- Present findings in an interactive dashboard filterable by drug, demographics, time, and severity—no medical jargon required.

We use Apache Spark for processing, Louvain community detection for drug clustering, XGBoost/LightGBM for severity prediction, and D3.js for interactive network visualization. This combination has not been applied to the full FAERS dataset before. The platform targets patients, clinicians, FDA researchers, pharma companies, and public health agencies.

2. Problem Definition

Current tools cannot handle the full FAERS volume interactively, lack predictive capabilities, and offer no network-based views. We address this with three components:

- **Graph construction:** Bipartite drug–adverse-event graph with community detection to reveal drug clusters with similar risk profiles.
- **Severity prediction:** Model to predict serious outcomes (hospitalization, death) from temporal, demographic, and network-derived features.
- **Interactive visualization:** Dashboard for exploring the graph, predictions, and temporal trends with real-time filtering.

Key constraints: data quality (duplicates, inconsistent naming), class imbalance (~38% serious), and visual scalability for networks with thousands of nodes.

3. Literature Survey

Pharmacovigilance & signal detection. Bate & Evans [1] establish disproportionality baselines (PRR, ROR, BCPNN) we compare against; purely statistical, ignoring temporal/network structure. Harpaz et al. [2] detect multi-drug interaction signals in FAERS but with high computational cost and no interactive interface. Sakaeda et al. [3] validate population-level FAERS mining but without visualization or predictive models.

Graph-based drug analytics. Zitnik et al. [4] (Decagon) use graph convolutional networks for polypharmacy side effects but require protein-interaction data absent from FAERS. Ferdousi et al.

Questions assigned to the following page: 1 and 2

[5] validate network topology features as drug interaction predictors on a small dataset. Tatonetti et al. [6] build the OFFSIDES/TWOSIDES ground-truth databases we use for validation.

Interactive health visualization. Chau et al. [7] (Apolo) introduce user-driven graph exploration informing our dashboard design. Plaisant et al. [8] (LifeLines) offer temporal visualization patterns for medical data. Gotz & Borland [9] survey healthcare visualization guidelines that shape our dashboard architecture.

4. Proposed Method

4.1 Data Acquisition and Preprocessing

We downloaded FAERS quarterly files (2004 Q1–2025 Q4) and processed them with Apache Spark on the GT PACE cluster:

- **Deduplication:** Using CASENUM/CASEVERSION, reduced ~35M raw records to ~28M unique cases. Further reduction is possible via fuzzy matching on demographics and narratives, which we plan to explore.
- **Drug name standardization:** Mapped to RxNorm identifiers (RxCUI), resolving brand-to-generic mappings; ~92% of entries normalized.
- **Event standardization:** Adverse event terms mapped to MedDRA preferred terms.
- **Severity encoding:** Reports with hospitalization, life-threatening events, disability, or death labeled serious (~38%). Missing demographics handled via indicator-variable imputation.

4.2 Network Construction and Community Detection

We built a weighted bipartite graph G with ~12K drug nodes and ~18K adverse event nodes, edges weighted by co-occurrence count. We projected G onto a drug–drug co-occurrence network G_D (edges where drugs co-appear in ≥5 reports; ~12K nodes, ~180K edges) and applied Louvain community detection with resolution tuned via modularity maximization:

- 47 communities detected; 31/47 (66%) have a dominant ATC drug class (>50% of members), e.g., SSRI cluster sharing mood-related events, fluoroquinolone cluster sharing tendon/neuropathy events.
- Several mixed-class communities share unexpected adverse event profiles—candidates for novel interaction signals.
- Computed centrality metrics (degree, betweenness, PageRank) per drug—used as ML features and dashboard visual encodings.

4.3 Machine Learning for Severity Prediction

We trained XGBoost and LightGBM to predict serious outcomes. Features organized into four groups:

- Drug-level: concomitant drug count, Louvain community ID, degree/betweenness/PageRank centrality.
- Event-level: number of reported events, flags for known high-risk MedDRA terms.
- Demographic: patient age (binned), sex, weight, reporter type.

Questions assigned to the following page: 1 and 2

- Temporal: reporting quarter, days since drug approval, rolling 12-quarter reporting rate per drug–event pair.

Temporal split (train: 2004–2022, val: 2023, test: 2024–2025) prevents leakage. We used class-weighted training over SMOTE after finding SMOTE degraded calibration. Preliminary validation results:

- LightGBM: AUC-ROC 0.81, F1 0.74, Precision 0.71, Recall 0.77
- XGBoost: AUC-ROC 0.79, F1 0.72, Precision 0.69, Recall 0.75
- PRR logistic regression baseline: AUC-ROC 0.67, F1 0.59

SHAP analysis confirms top features: concomitant drug count, patient age, reporter type, and betweenness centrality—validating that network features carry predictive signal. Optuna tuning (200 trials) is in progress.

4.4 Interactive Visualization Dashboard

D3.js dashboard with four coordinated, cross-filtered views:

- **Network view:** Force-directed drug co-occurrence graph. Node size = degree, color = community, edge thickness = weight. Level-of-detail rendering (cluster supernodes at low zoom, individual drugs at high zoom). Renders ~5K nodes at 30+ FPS.
- **Temporal trend chart:** Reporting rates over time for selected drugs, with anomaly highlighting (>2σ from rolling mean).
- **Demographic panel:** Age/sex distribution of adverse events for selected drugs or clusters.
- **Severity prediction panel:** Predicted serious-outcome probability for user-specified drug–demographic combinations, with SHAP waterfall explanations.

All views are linked via cross-filtering. Remaining work: drug search bar, severity filter toggles, CSV export.

4.5 Innovations

- **Graph-derived features for severity prediction:** Network centrality and community membership as ML features—novel in pharmacovigilance.
- **Scalable interactive FAERS network visualization:** Level-of-detail rendering for the full dataset; no existing public tool offers this.
- **Coordinated multi-view dashboard with ML insights:** Cross-filtered views with SHAP explanations.
- **Community detection for drug risk profiling:** Louvain clustering to discover unexpected drug groupings and interaction signals.

5. Experiments and Evaluation

5.1 Completed

- **Severity prediction:** LightGBM (AUC 0.81) outperforms XGBoost (0.79) and PRR baseline (0.67). Ablation removing network features drops AUC to 0.77.
- **Community validation:** 31/47 communities (66%) align with ATC drug classes; 4 mixed-class communities under investigation as novel interaction signals.

Questions assigned to the following page: 1 and 2

- **Dashboard prototype:** Four coordinated views functional with cross-filtering; network view handles ~5K nodes at 30+ FPS.

5.2 Planned

- **FDA alert validation:** Compare flagged pairs against FDA safety communications (2020–2025); measure recall and temporal lead time.
- **Scalability:** Wall-clock time for Spark pipeline at 10–100% data volume; dashboard FPS vs. node count.
- **User study:** 5–10 participants, three tasks (top events, similar-risk drugs, severity assessment); collect time, accuracy, feedback.
- **Final evaluation:** AUC-ROC, F1, precision, recall, calibration curves, SHAP plots on 2024–2025 test set.

6. Conclusions (Preliminary)

Progress confirms feasibility: Spark handles 30M+ records, the drug network shows meaningful community structure, and ML models outperform baselines with clear network feature contribution. The four-view dashboard is functional. Remaining: hyperparameter tuning, FDA alert validation, scalability benchmarks, user study, and dashboard polish. We are on track for the final deadline.

Risks: Data quality—mitigated via deduplication/RxNorm; class imbalance—class-weighted training; visual clutter—level-of-detail rendering. **Cost:** $0. All data (FAERS), compute (personal machines + GT PACE), and software are free/open-source.

7. Plan of Activities

Consolidated original and revised timelines below. Highlighted rows changed: replaced SMOTE with class-weighted training (effort to Optuna/SHAP); started FDA validation one week earlier.

Activity	Old Start	Old End	New Start	New End	Owner
Proposal & literature survey	1	2	1	2	All
FAERS acquisition & Spark cleaning	2	4	2	4	Pradeep
Drug name standardization (RxNorm)	3	4	3	4	Sharath
Network constr. & community det.	4	6	4	6	Pradeep, Sunil
ML baseline models	4	6	4	6	Sharath
Prototype interactive viz (D3.js)	5	7	5	7	Sunil
Midterm / progress report	6	6	6	6	All
Hyperparameter tuning & SHAP	7	9	7	9	Pradeep, Sharath
Dashboard polish: search, export	7	10	7	10	Sunil
Validation against FDA alerts	9	10	8	10	Sharath
Scalability experiments	10	11	9	11	Pradeep
User study & evaluation	10	11	10	11	Sunil
Final report & poster	11	12	11	12	All

Questions assigned to the following page: 1 and 2

Effort distribution: All three team members have contributed roughly equally to the project so far. Each member owned distinct components—data pipeline, ML modeling, and visualization—while collaborating on the literature survey, integration testing, and this report.

References

[1] A. Bate and S. Evans. Quantitative signal detection using spontaneous ADR reporting. Pharmacoepidemiol. Drug Saf., 18(6):427–436, 2009.

[2] R. Harpaz, H. S. Chase, and C. Friedman. Mining multi-item drug adverse effect associations. BMC Bioinformatics, 11(S9):S7, 2010.

[3] T. Sakaeda et al. Data mining of the public FDA adverse event reporting system. Int. J. Med. Sci., 10(7):796–803, 2013.

[4] M. Zitnik, M. Agrawal, and J. Leskovec. Modeling polypharmacy side effects with graph convolutional networks. Bioinformatics, 34(13):i457–i466, 2018.

[5] R. Ferdousi, R. Safdari, and Y. Omidi. Computational prediction of drug-drug interactions. J. Biomed. Inform., 70:54–64, 2017.

[6] N. P. Tatonetti et al. Data-driven prediction of drug effects and interactions. Sci. Transl. Med., 4(125):125ra31, 2012.

[7] D. H. Chau et al. Apolo: Making sense of large network data. Proc. CHI, pp. 167–176, 2011.

[8] C. Plaisant et al. LifeLines: Visualizing personal histories. Proc. CHI, pp. 221–227, 1996.

[9] D. Gotz and D. Borland. Data-driven healthcare: Challenges for interactive visualization. IEEE CG&A, 36(3):90–96, 2016.